

B. Tech Degree V Semester Examination, November 2009

ME 504 THERMAL ENGINEERING

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Use of steam tables and Mollier charts permitted)

PART A

(Answer ALL questions)

(8 x 5 = 40)

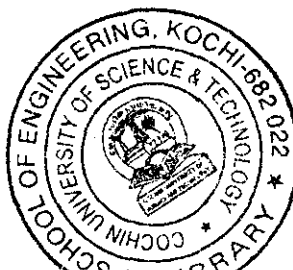
- I. (a) Derive an expression for the air-standard efficiency of Otto cycle.
(b) Define :
(i) Volumetric efficiency
(ii) Mean Effective Pressure
(iii) Relative Efficiency.
(c) What is Turbo charging? What is its effect on volumetric efficiency?
(d) Write notes on :
(i) Octane number
(ii) Cetane number
(e) What is critical pressure ratio? Explain the effect of friction on it.
(f) Explain velocity compounding of impulse turbines.
(g) Explain a gas turbine cycle with regeneration.
(h) Derive an expression for the efficiency of a roots blower.

PART B

(4 x 15 = 60)

- II. (a) What is meant by Indicator diagram? Draw and explain the indicator diagram of a four – stroke petrol engine. (6)
(b) A four – stroke petrol engine delivers a brake power of 36.8kW with a mechanical efficiency of 0.8. The A/F ratio is 15 : 1 and the fuel consumption is 0.4068 kg/kWh. The heating value of the fuel is 4200kJ/kg. Calculate :
(i) Indicated power
(ii) Frictional power
(iv) Brake and indicated thermal efficiencies
(v) Total fuel consumption
(vi) Air consumption per second. (9)
- OR**
- III. (a) What is meant by heat balance sheet? Explain the procedure for preparing it. (6)
(b) A four cylinder four – stroke petrol engine having a bore of 6.5 cm, a stroke of 9.5 cm, speed 3000 rpm and clearance volume 65 cm³. Determine the brake mean effective pressure and SFC. Given relative efficiency = 0.5 (Based on the brake thermal efficiency) Heating value of fuel = 4200kJ/kg. The engine developed 68.6 N-m torque when tested on load. (9)
- IV. (a) Explain with neat diagram, the magneto ignition system. (6)
(b) Explain the combustion phenomenon of an S I Engine. (9)
- OR**
- V. (a) Explain the working of Wankel engine. (6)
(b) Explain the various designs of combustion chambers for C I Engines. (9)

(Turn Over)



- VI. (a) Explain super saturated flow through nozzles. (6)
 (b) A steam nozzle is supplied with steam at 7 bar and 250°C and it discharges steam at 1 bar. If the delivering portion of the nozzle is 50 mm long and the throat diameter is 5 mm, determine the cone angle of the divergent portion. Assume 10% of the total available entropy drop to be lost in friction at the diverging part. Also determine the velocity and temperature of steam at the throat. (9)
- OR**
- VII. (a) Explain degree of reaction of a steam turbine. (5)
 (b) A single stage impulse turbine rotor has a diameter of 1.2m running at 3000 rpm. The nozzle angle is 18° and blade speed ratio is 0.42. The ratio of relative velocities at outlet to inlet is 0.9. The blade outlet angle is 3° smaller than inlet angle. If the steam flow rate is 5 kg/sec, draw the velocity diagram and find the following : (10)
- (i) Velocity of whirl
 - (ii) Axial thrust on the end bearing
 - (iii) Blade angles
 - (iv) Power developed
- VIII. The following data refer to a gas turbine having two stages with inter cooling and reheat : Pressure ratio = 4, $T_{\min} = 27^{\circ}\text{C}$, $T_{\max} = 600^{\circ}\text{C}$, Turbine efficiency = 0.8, Compressor efficiency = 0.8, Regenerator effectiveness = 0.8. Determine cycle thermal efficiency, Cycle work ratio and cycle air rate. Assume pressure ratio to be same for both the stages and intake pressure to compressor as 1 bar $C_p = 1.0\text{kJ/kgK}$. (15)
- OR**
- IX. (a) Draw and explain the typical velocity diagrams of a rotodynamic compressor. (6)
 (b) A rotary vane compressor works between the pressure limits of 1 bar and 1.5 bar and gives $4\text{ m}^3/\text{min}$ of free air delivered when running at 200 rpm. Determine the power required to drive the compressor when
- (i) the ports are so placed that there is no internal compression
 - (ii) the ports are so placed that there is 50% pressure rise due to internal adiabatic compression before backflow occurs.
- Determine also the blower efficiency. Take mechanical efficiency as 100%. (9)
